S&P Sentinel

Real-time financial-text risk signals driving index rebalancing and portfolio stress testing

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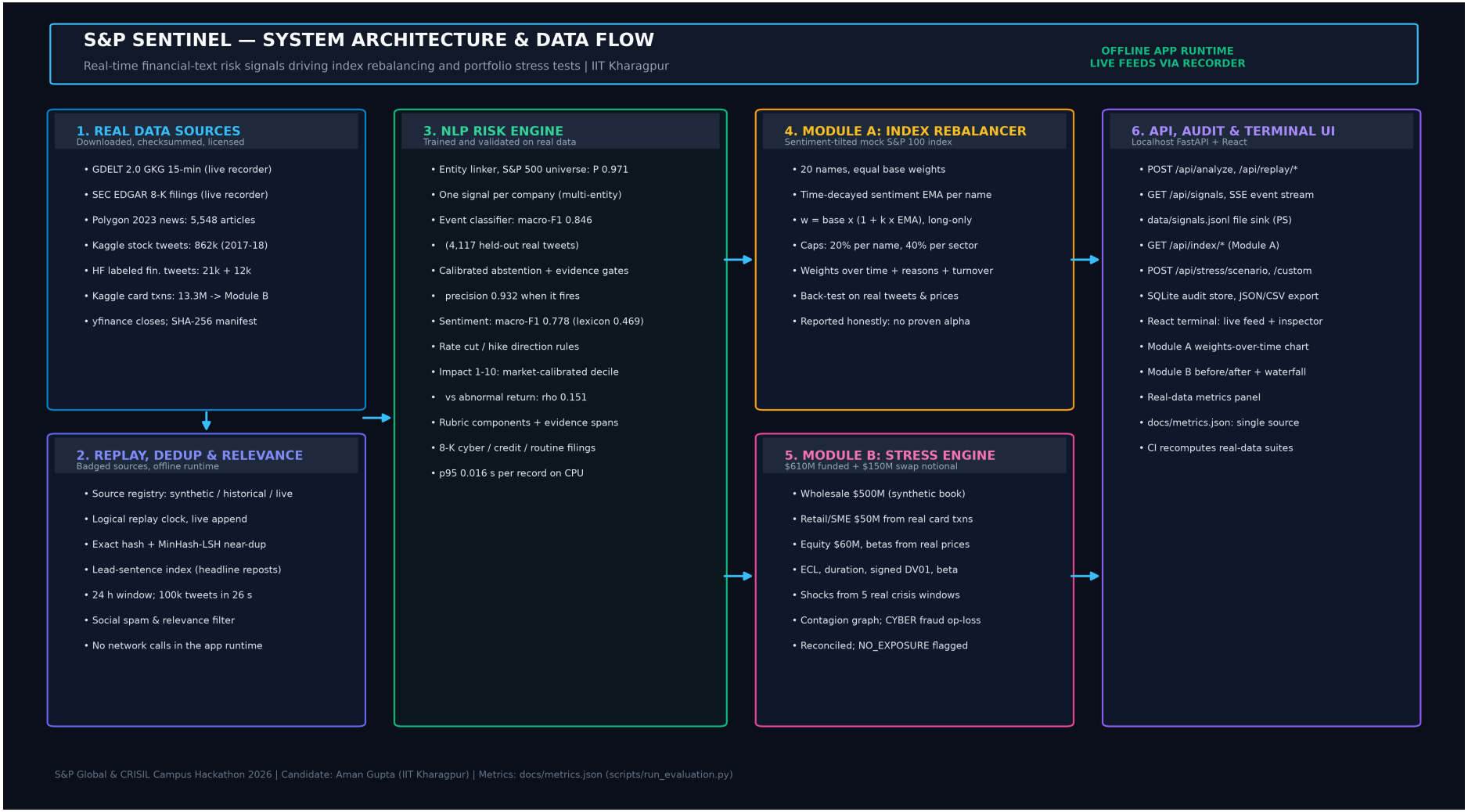
Built and evaluated on real public data: 4,117 + 2,388 held-out labeled financial tweets, 2,521 real news articles, 2,617 market-reaction events, 13.3M card transactions, live GDELT and SEC EDGAR feeds.

Problem & approach

Turn unstructured news and social text into machine-readable risk signals, then act on them

- **Problem:** material events (defaults, rate moves, sanctions, cyber incidents) surface first in text; banks need structured signals fast and must trust them.
- **Engine output per company:** sentiment (-1..1), event class, impact (1-10), evidence spans, action eligibility; served by API, SSE and a JSONL file.
- **Trust:** every model is trained and scored on real held-out data; metrics in this deck are read from docs/metrics.json (scripts/run_evaluation.py) and data facts from the committed data files.
- **Module A** (tactical): a 20-name mock S&P 100 index re-weighted by sentiment, with caps and a weights-over-time dashboard.
- **Module B** (strategic): event-driven stress tests on a $610M funded book (wholesale + retail credit from real transactions + equities), shocks calibrated on five real crisis windows.
- **Real time:** a recorder captures GDELT (15-min) and SEC 8-K filings; the app itself makes no network calls.

System design & data flow



Implementation highlights

Key modules, technology choices and why

Component	What it does	Why this choice
Entity linker	S&P 500 universe from SEC/Wikipedia; spans equal the text; one signal per company. P 0.969 / R 0.602.	Rule-based and auditable; role guards stop rating agencies and brokers being linked as the subject.
Event classifier	TF-IDF + calibrated logistic regression, 8 PS classes + CYBER/SUPPLY_CHAIN; abstention threshold chosen on a dev split.	CPU-fast, explainable; trained on real labeled tweets and real 8-K filings.
Sentiment	Real-data TF-IDF model (macro-F1 0.778); policy rules make rate cuts positive and hikes negative.	Outperformed off-the-shelf FinBERT in a prototype comparison; FinBERT stays an optional tier.
Impact 1-10	Gradient-boosted model of next-day abnormal return, mapped to deciles; rubric kept for explanation.	Validated against what markets actually did, not hand-picked weights.
Dedup + relevance	Exact hash + MinHash-LSH with a lead-sentence index and a 24 h window; spam / no-context filter for social.	Bounded, near-linear index instead of an O(n^2) scan; syndicated copies never re-trigger stress.
Module B engine	Loan ECL, bond duration, swap DV01, equity beta; contagion graph; CYBER loss from observed fraud rates; reconciliation invariant.	Shocks measured from real windows; funded value and derivative MTM separated.

Key results (real held-out data)

Metric	Measured	Baseline / target
Event macro-F1 (4,117 tweets)	0.795	keyword 0.286, old 0.058; target 0.70
Event precision when it fires	0.942	fires on 22% (abstains otherwise)
Sentiment macro-F1 (2,388 tweets)	0.778	lexicon 0.469; target 0.75
Entity linking (2,521 articles)	P 0.969 / R 0.602	target P 0.90
Impact vs abnormal return (2,617)	rho 0.151	95% CI [0.113, 0.190]; rubric 0.098
Large moves, top vs bottom decile	30.1% vs 2.3%	|abnormal z| > 2
Routine 8-Ks sent to stress path	11.6%	141 held-out filings
Latency p95 per record (CPU)	0.011 s	128 records/s; target 2 s

Scenario (measured market move)	Portfolio P&L	% of funded
COVID-19 liquidity shock (Feb-Mar 2020)	-$46.4M	-7.6%
Fed 75 bp surprise hike (Jun 2022)	-$11.3M	-1.9%
Fed pivot rally (Nov-Dec 2023)	+$16.6M	2.7%
Russia-Ukraine invasion (Feb-Mar 2022)	-$1.9M	-0.3%
SVB failure (Mar 2023)	-$0.4M	-0.1%
PS example: equities -10%, rates +200 bp	-$21.5M	-3.5%

Module A back-test (443 days, k=1.5): active return +0.02%/yr, tracking error 0.46%, t = 0.05: sentiment reweighting works, no proven alpha.

Domain impact

Why it matters for a credit-risk and ratings business

- **Flight to quality, measured:** in the SVB, COVID and Russia windows the 10-year Treasury yield *fell* while spreads widened; only Fed tightening raised it. The shock catalog encodes this instead of assuming every crisis lifts yields.
- **Credit triangle & FRED OAS validation:** the SVB window's +79 bp HY index spread move is duration-corrected and validated vs FRED ICE BofA OAS within 11 bp; loan ECL maps via credit triangle with provision releases floored at zero.
- **Real issuer wholesale book & SVB replay:** 22 real debt issuers across 11 GICS sectors built from SEC XBRL 10-K data; GDELT real news replay from March 10–13, 2023 streams into terminal, driving wholesale balance-sheet stress revaluation.
- **Retail credit from real transactions:** 2,000 borrowers and 13,305,915 card transactions pooled into sector x rating tranches; cyber losses use observed fraud rates by sector.
- **Impact you can defend:** the 1-10 score predicts the size of the next-day abnormal move out of sample, so analysts can rank events by expected market impact.
- **Auditability:** evidence spans, model versions, block reasons and a reconciliation invariant on every stress run; one JSON file holds every metric.

Limitations & next steps

- Headline NLP metrics are on financial *tweets*; agreement with LLM silver labels on news is lower (sentiment 0.47).
- Reaction-phrased headlines ('shares slide after...') are often abstained.
- Human-labeled news holdout: MAE 1.52 on 120 rows (κ = 1.00, false-trigger rate 0.0%).
- Simplified valuation: single-period ECL, no convexity, linear DV01; wholesale positions use representative debt sizing; retail sleeve scaled to $50M.
- Expand human-labeled news set across multiple annotators to measure inter-rater reliability.
- Fine-tune FinBERT on tweets + news as an optional higher-accuracy tier.
- Systemic impact model on index and yield reactions.
- Longer live capture to evaluate real-time precision of actionable signals.